

Foreword to the Second Edition

Much has changed in the world of fractals, computer graphics and modern mathematics since the first edition of *Fractals Everywhere* appeared. The company Iterated Systems, Inc., founded by Michael Barnsley and Alan Sloan, is now competing in the image compression field with both hardware and software products that use fractal geometry to compress images. Indeed, there is now a plethora of texts on subjects like fractals and chaos, and these terms are rapidly becoming "household words."

The fundamental approach to fractal geometry through iterated function systems remains sound as an introduction to the subject. This edition of *Fractals Everywhere* leaves this approach largely as it stands. One still needs a grounding in concepts in metric space theory and eventually (see Chapter IX) measure theory to get a working understanding of the subject. However, there have been several additions to help ease and broaden the reader's development.

Primary to these is the addition of answers to the mathematical problems. These were done largely by starting at one end of the book writing the answers until the other cover was reached. Most of the answers found in the key have been worked over at least twice, in hopes improving the accuracy of the key. Every effort has been made to rely solely on the material presented ahead of each problem, although in a few of the harder problems some concepts have been introduced in the answers themselves. These are not considered necessary to the development of the main thread of the text; however, if the reader finds some areas of mathematics touched on in looking at the presented solutions which extend the feeling for the subject, the key has served its purpose.

In addition to the answer key, there have been some other changes as well. In Chapter III, section 11, the main theorem has been qualified. The reader with more mathematical background will recognize that the additional Lipschitz condition satisfies the need for equicontinuity in Theorem 11.1. This is not the only way to satisfy it, just the clearest in terms of the presumed mathematical background.

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There have been problems added to several chapters to develop the idea of Cartesian products of code spaces. This was done because it helps bridge the gap between IFS theory and the reversible systems found in physical chaos, and because it presents an interesting way of looking at the Random Iteration Algorithm in Chapter IX. The thread of these problems begins in Chapter II, leads up to the baker's transformation in Chapter IV, and is completed as an example in Chapter IX. Additional problems were added in Chapter III to develop some basic properties of eigenvalues and eigenvectors, which can be useful in examining dynamics both from the point of view described in the text, and elsewhere. It is hoped that with these additional tools those readers whose goals are application-oriented will come away with more at their disposal, while the text itself will retain its readable style.

I would like to thank Lyman Hurd for many useful discussions about the topological nature of nonempty compact sets, and John Elton for his patience while I ran many of my new examples and problems past him to check them and to check the "excitement level" of the additional material.

Hawley Rising

It seems now that deterministic fractal geometry is racing ahead into the serious engineering phase. Commercial applications have emerged in the areas of image compression, video compression, computer graphics, and education. This is good because it authenticates once again the importance of the work of mathematicians. However, sometimes mathematicians lose interest in wonderful areas once scientists and engineers seem to have the subject under control. But there is so much more mathematics to be done. What is a useful metric for studying the contractivity of the vector recurrent IFS of affine maps in $\mathbb{R}^2$? What is the information content of a picture? Measures, pictures, dreams, chaos, flowers and information theory—the hours of the days keep rushing by: do not let the beauty of all these things pass us by too.

Michael Fielding Barnsley